

SE 410 – SOFTWARE FRAMEWORK APPLICATIONS
Spring 2025-2026

E-Learning Platform with Personalized Study Plans
Project Requirements



Group 7

Defne Yılmaz
Hasan Efe Ünal
Can Özer
Ege Yılmaz

Functional Requirements

1. The system should allow admins to create new courses by entering details such as title, description, category, difficulty level, and ECTS credit.
2. The system should allow admins to edit or delete existing courses in the course catalog.
3. The system should allow students to browse and view all available courses on the web platform.
4. The system should allow students to request enrollment in a course, and the enrollment must be approved by an admin before the student gains access.
5. The system should allow students to access course modules and mark them as completed.
6. The system should track students' progress for each enrolled course and display the completion percentage.
7. The system should generate personalized study plans based on students' progress, enrolled courses, and ECTS credit load to ensure balanced workload.
8. The system should recommend courses to students using delegates based on their learning preferences, performance, and past enrollments.
9. The system should allow admins to view and monitor student progress through the desktop application.
10. The system should allow admins to analyze course-related data such as enrollment numbers, completion rates, and student activity metrics.

Non-Functional Requirements

1. The system should provide a consistent and intuitive user interface across all pages, including clear navigation structures, readable layouts, and feedback messages to improve ease of use.
2. The system should ensure efficient and secure data handling by applying password hashing and encryption mechanisms for sensitive information.
3. The system should support two-factor authentication (2FA) to enhance user account security.

4. The system should use a shared SQLite database and ensure that all data remains consistent and synchronized across both desktop and web applications.
5. The system should include proper documentation and a user manual to guide users in effectively using the platform.